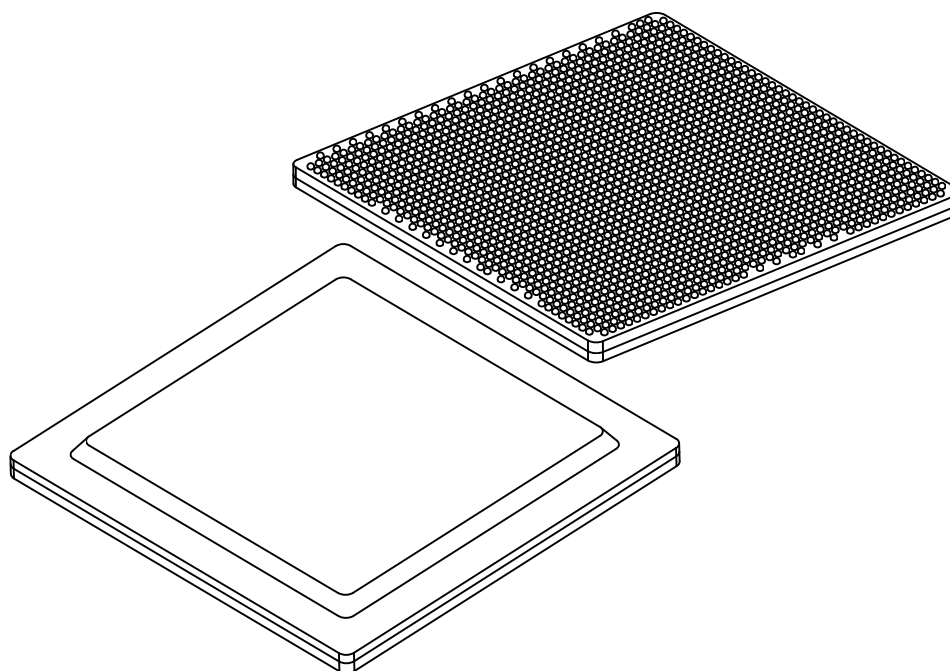


Packaging Diagrams and Parameters

1894-Ball Thick Ball Grid Array-Flip Chip (7JC) 45x45x3.90mm Body [BBGA] With Heat Spreader

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	1894		
Pitch	e	1.00 BSC		
Overall Height	A	3.34	3.62	3.90
Ball Height	A1	0.40	0.50	0.60
Substrate Thickness	A2	1.17 REF		
Package Thickness	A3	2.94	3.12	3.30
Overall Length	D	45.00 BSC		
Ball Array Length	D1	43.00 BSC		
Top of Heat Spreader Length	D2	37.00 BSC		
Overall Width	E	45.00 BSC		
Ball Array Width	E1	43.00 BSC		
Top of Heat Spreader Width	E2	37.00 BSC		
Ball Diameter	b	0.50	0.64	0.70

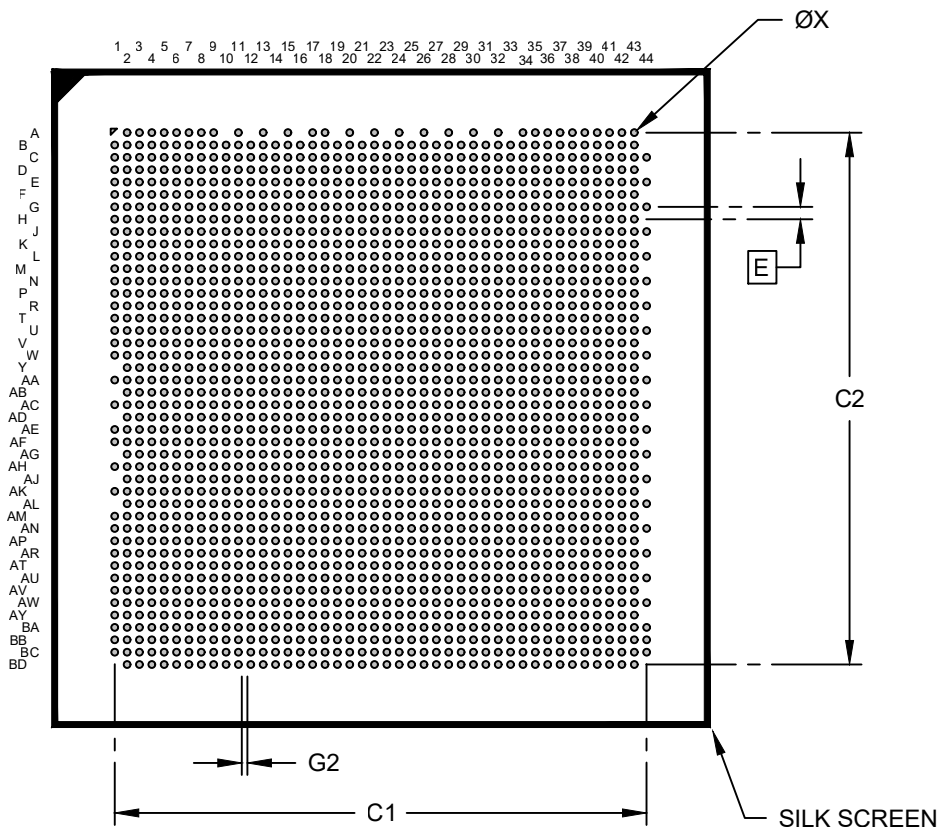
Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

1894-Ball Thick Ball Grid Array-Flip Chip (7JC) 45x45x3.90mm Body [BBGA] With Heat Spreader

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Contact Pitch	E		1.00 BSC	
Contact Pad Spacing	C1		43.00	
Contact Pad Spacing	C2		31.00	
Contact Pad Diameter (X1894)	X			0.55
Contact Pad to Contact Pad	G2	0.45		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, please refer to current industry standard IPC-7093.